9 Fig. 9.1 shows a circuit with an alternating current (a.c.) supply, a resistor and a diode.

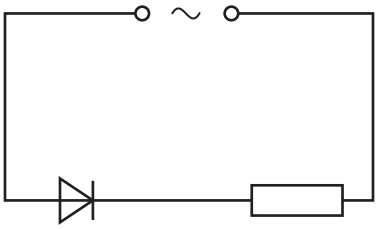


Fig. 9.1

The frequency of the power supply is 50 Hz.

(a) Calculate the time period (time for one complete cycle) of the a.c. supply.

time = ... [2]

(b) The peak potential difference (p.d.) across the resistor is 340 V.

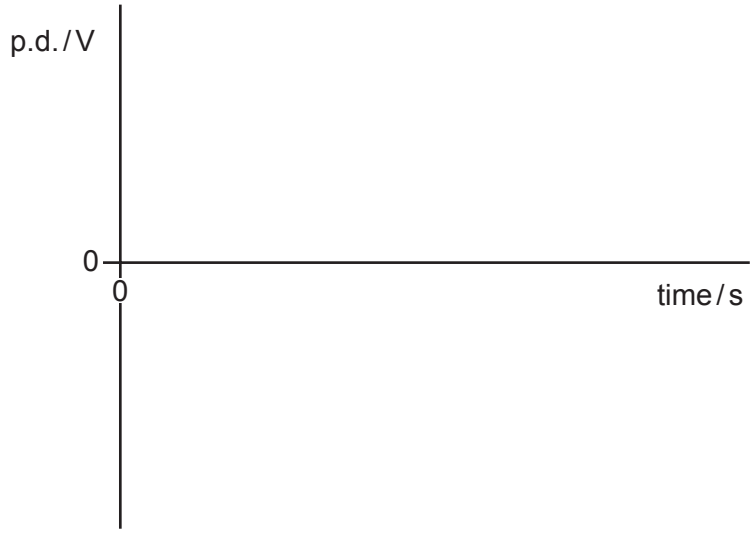


Fig. 9.2

On Fig. 9.2:

(i) sketch a graph to show how the p.d. across the resistor varies with time for two cycles [2]

(ii) label the p.d. axis with the value of p.d. at the peak [1]

(iii) label the time axis with two values of time. [2]

[Total: 7]